Partnership and Alligations & Mixtures

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↳ **Partnership:-**

When there is no sufficient money to start a Business by a Investor then he needs the support of partners to start the business. This is called 'Partnership Business'.

↳ **Kind of Partners:-**

Partners are of 2 types:-

(i) Active Partner. (ii) Sleeping Partner.

(i) **Active Partner:-** Investor, who is investing in the Business as well as actively participating in the Business processings is "Active Partner".

(ii) **Sleeping Partner:-** Investor, who is simply investing but not participating actively in the Business Processings is "Sleeping Partner".

↳ **Kind of Partnerships:-** Partnerships are of 2 types:-

(i) Simple Partnership (ii) Compound Partnership.

(i) **Simple Partnership:-** In simple Partnership time period of investment in the Business is constant irrespective of their investment of Partners.

In Simple Partnership Profit/Loss Ratio of Partners is.

P/L Ratio = Invest of Part 1 : Invest of Part 2 : . . . .

$= I_1 : I_2 : \ldots$ [Because Time Period is not considerable]

(ii) Complex / Compound Partnership:-

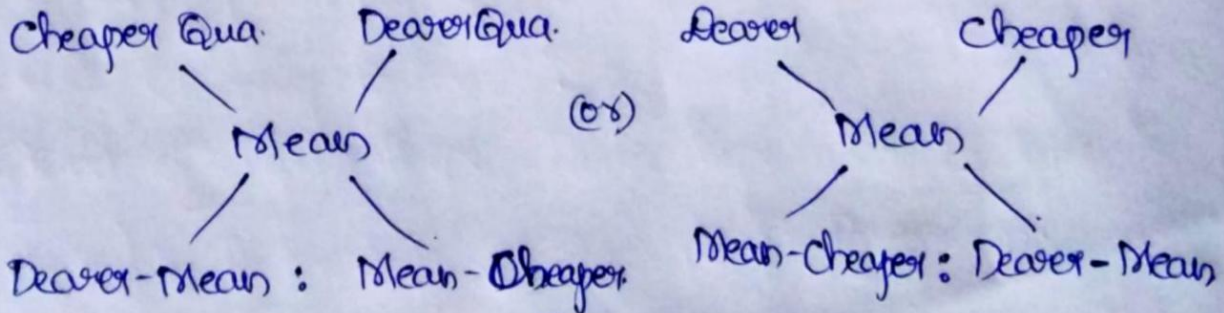
The time period of Investment of Investors in the Business is different irrespective of Investment of Investors.

In Complex Partnership Profit/Loss Ratio of Partners is. P/L Ratio = $I_1 \times$ timePeriod 1 : $I_2 \times$ T.P2 : $I_3 \times$ T.P3 . . .

[Time period is Considered because it's different].

2) Alligations & Mixtures:-

When 2 or more different valued articles are there, then to produce new valued quantity which is in between these valued quantity, Alligations & Mixtures procedure is to be used.



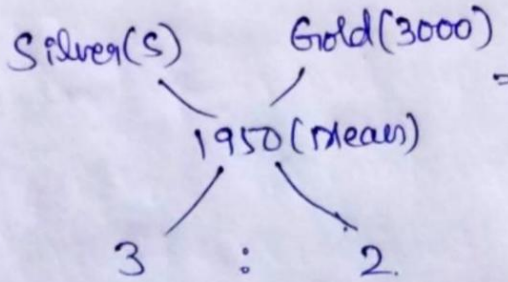
Alligations Mixtures Ratio = $\frac{\text{[Dearer - Mean] Prices}}{\text{[Mean - Cheaper] Prices}}$ (or)

$= \frac{\text{Mean - Cheaper}}{\text{Dearer - Mean}}$

Explanation to Booklet Questions:-

Ex:-1:- [B.Q:-1]:- A jeweller mixes silver & Gold in the ratio 3:2 in order to make a necklace. The cost of Gold is 3000/- per gram. If the cost of a gram of mix is 1950/-. The Price of Silver per gram is?

Ans: $\frac{\text{Quant of Cheaper Ingradient}}{\text{Quant of dearer Ingradient}} = \frac{3}{2} = \frac{D-M}{M-C}$



$\Rightarrow \frac{\text{Gold-Mean}}{\text{Mean-Silver}} = \frac{3}{2}$

$\frac{3000-1950}{1950-S} = \frac{3}{2}$

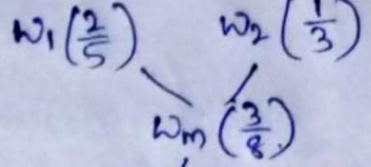
$\Rightarrow (1050)2 = 3(1950-S) \Rightarrow 2100 = 5850-3S$

$\Rightarrow 3S = 3750 \Rightarrow \boxed{S=1250}$

Ex:-2:- [B.Q:-2]:- 2 vessels contains water & Milk in the ratio 2:3 & 1:2. In which Ratio the contents of two vessels to be mixed. which contains the water & Milk in 3:5 Ratio?

Ans: Whenever the problems are given like this we have to take either milk (or) water fractions as Reference.

(i) Water Rat:-

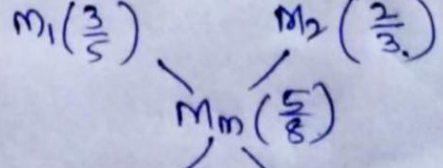


$\frac{3}{8}-\frac{1}{3} : \frac{2}{5}-\frac{3}{8}$

$\frac{9-8}{24} : \frac{16-15}{40} \Rightarrow \frac{1}{3} : \frac{1}{5} \Rightarrow 5:3$

(or)

(ii) Milk Rat



$\frac{2}{3}-\frac{5}{8} : \frac{5}{8}-\frac{3}{5}$

$\frac{16-15}{24} : \frac{25-24}{40} = \frac{1}{24} : \frac{1}{40}$

$= \frac{1}{3} : \frac{1}{5} \Rightarrow 5:3$